



**NATIONAL UNIVERSITY OF MODERN
LANGUAGES**

Human Computer Interaction Final Project Report

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Doctor Appointment Booking Web Application

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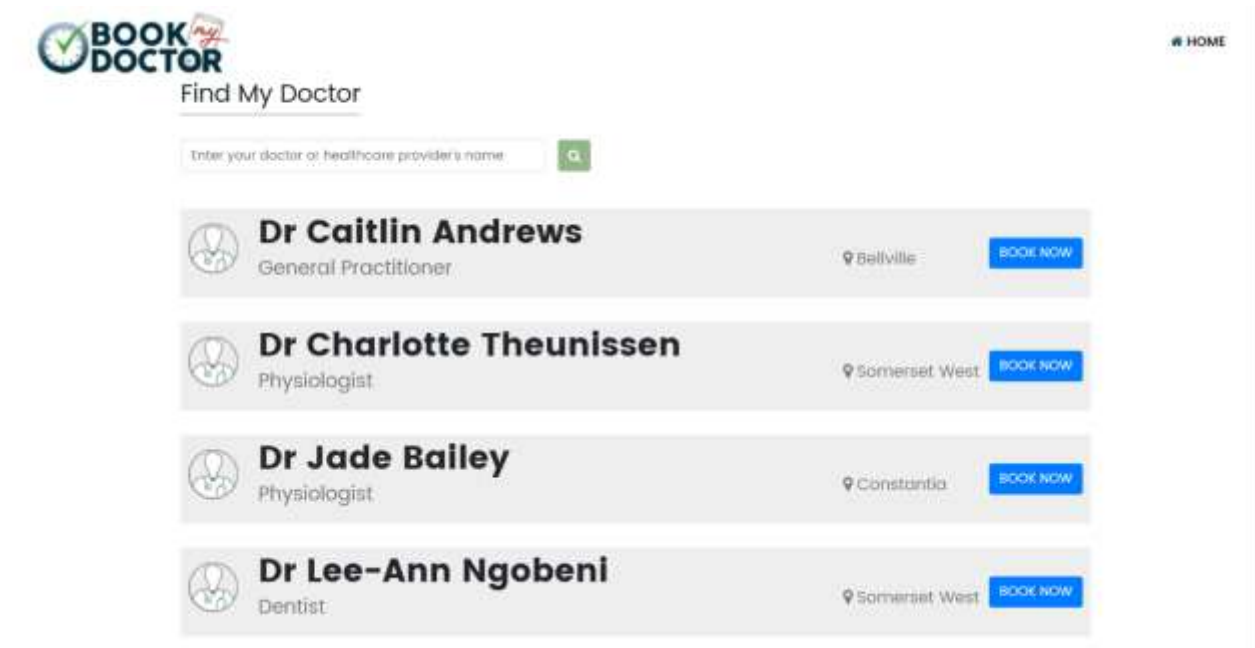


Figure 1 Existing Website Home Page



Figure 2 Existing Doctor Appointment Page



Your timeslot has been reserved for 5 minutes and will expire at 17:51. Please submit this form before the cut-off time.

Date : **Saturday, 27 December 2025**

Time : **09:00**

Main Member Information

Name*

Email*

Cell Number*

ID Number*

Medical Scheme

Medical Scheme Number

Patient Information

Figure 4 Existing Booking Page

Welcome Back

Login to continue to MediConnect

Login

or



Continue with Google

Don't have an account? [Sign Up](#)

Figure 3 New Login Page along with third party logins (Google)

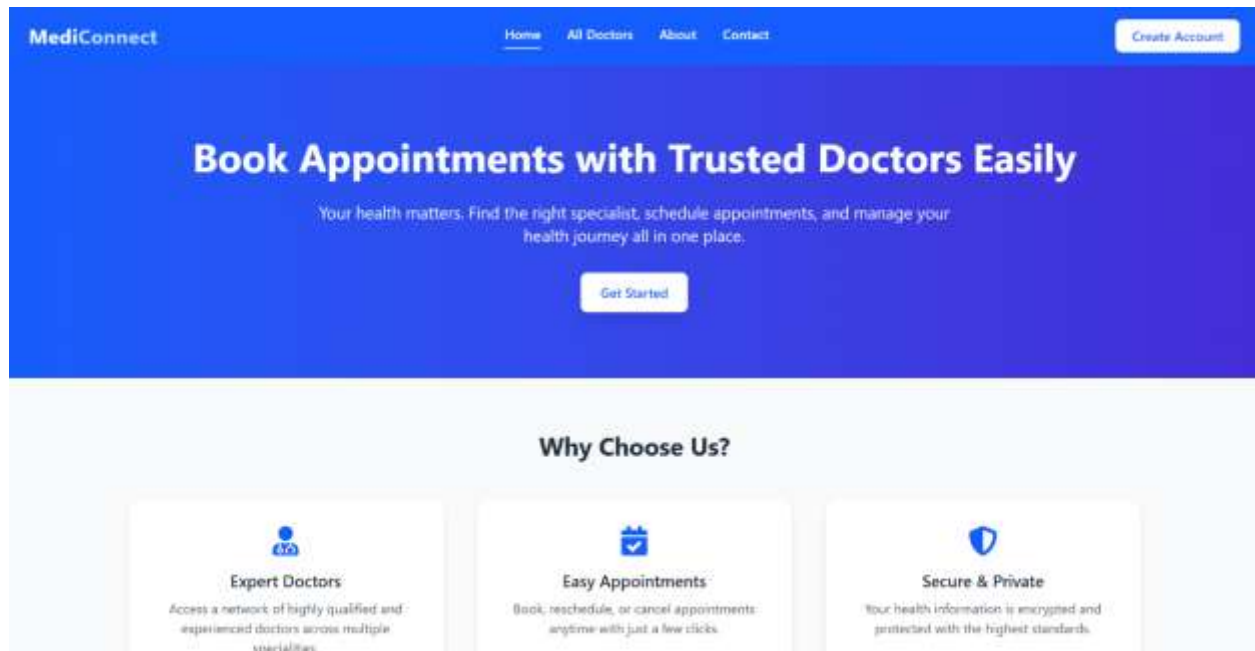


Figure 5 New Home Page redesigned

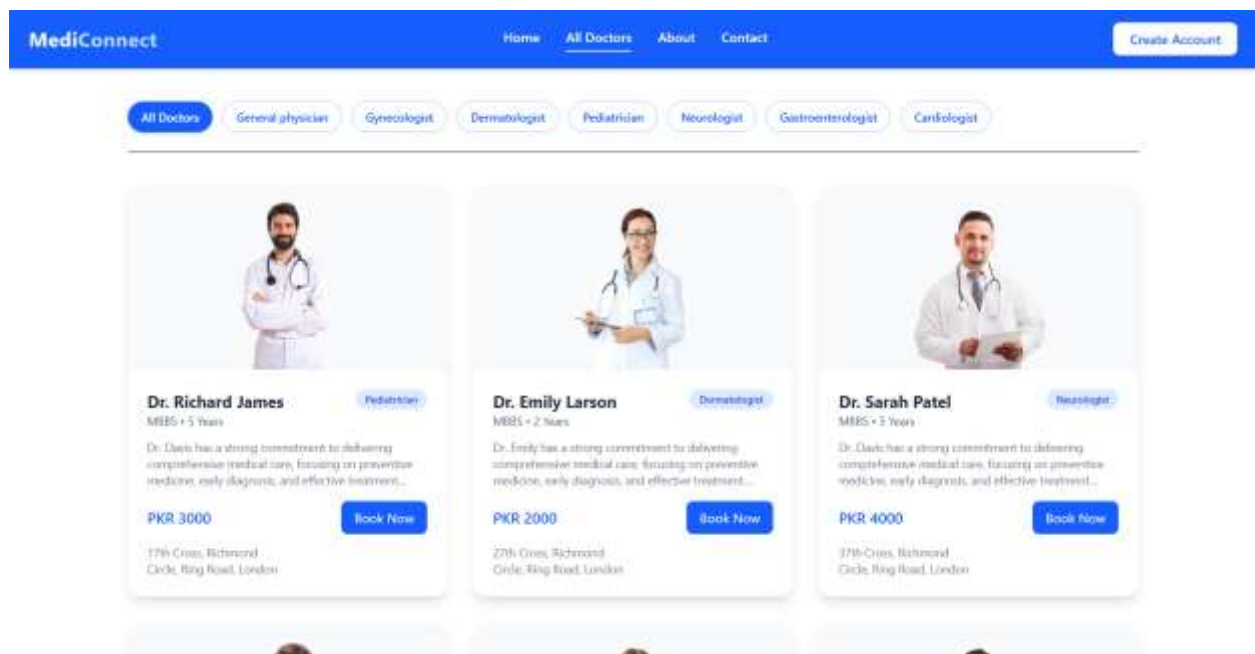


Figure 6 New Doctors Page

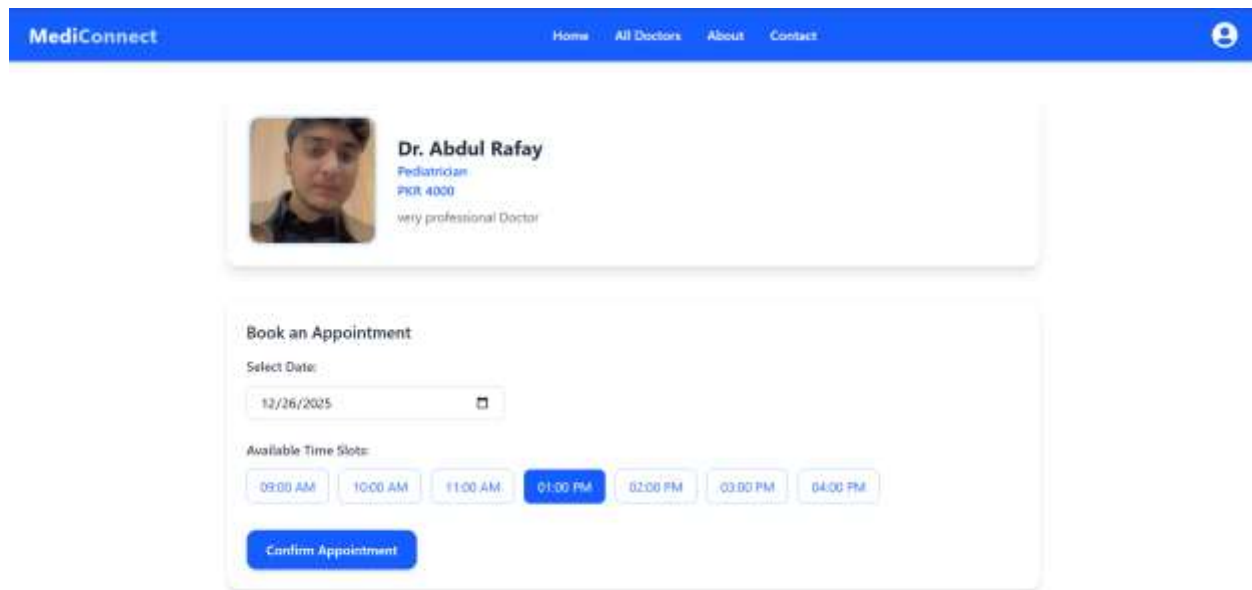


Figure 7 New Appointment Booking page With less complexity than existing one

1. Introduction

In today's digital era, healthcare systems are increasingly dependent on web-based applications to provide ease and accessibility to patients. One of the most critical healthcare services is **doctor appointment booking**, which must be simple, fast, and user-friendly.

For this semester project, we analyzed an existing doctor appointment booking website:

Original Website:

<https://www.bookmydoctor.co.za/find-my-doc/>

New project :

<https://github.com/git-rafaykhan/doctor-oppointment-booking.git>

The selected website suffers from **poor user interface design, confusing navigation, weak visual hierarchy, and usability issues**, making it difficult for users to book appointments efficiently.

To address these problems, our group **redesigned and rebuilt the system as a Doctor Appointment Booking Web Application**, focusing on:

- Improved UI/UX
- Better usability
- Clear navigation
- Enhanced user interaction

This report documents the **software requirements, usability problems, applied principles, graphical improvements, backend implementation, testing, and work distribution.**

2. Goals and Objectives

Goals

- To analyze a real-world healthcare web application with poor usability.
- To redesign the system using proper usability and UI/UX principles.
- To develop a working doctor appointment booking web application.
- To ensure ease of use for patients and doctors.

Objectives

- Identify usability and graphical issues in the existing website.
- Apply usability principles to improve efficiency and learnability.
- Implement frontend and backend functionality.
- Perform black box testing using automated tools.
- Deliver a complete, working prototype for demonstration.

3. Usability Issues of Existing Website

The existing website **bookmydoctor.co.za** presents several usability challenges:

Major Issues Identified

1. Complex and cluttered user interface.
2. Poor navigation structure.
3. Lack of visual hierarchy.
4. Difficult doctor search process.
5. No clear call-to-action buttons.
6. Inconsistent fonts and colors.
7. Mobile responsiveness issues.

Screenshot Placement

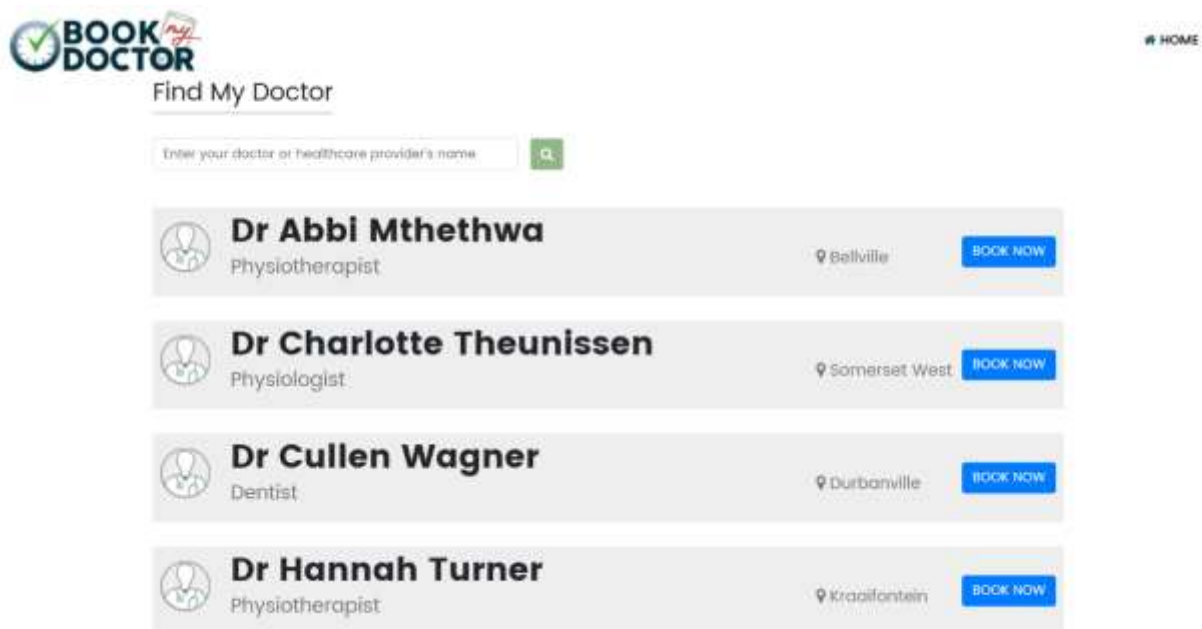


Figure 8 Existing Website Home Page



Figure 9 Existing Booking Page

4. Usability Principles Applied to Improve the Website (Metrics)

To improve the system, we applied the following usability principles:

1. Simplicity

- Reduced unnecessary text and components.
- Clear and minimal design layout.

2. Consistency

- Uniform color scheme and fonts.
- Consistent button styles and navigation menus.

3. Learnability

- Easy-to-understand flow for booking appointments.
- Icons and labels used clearly.

4. Efficiency

- Fewer steps required to book an appointment.
- Fast doctor search with filters.

5. Accessibility

- Readable font size.
- Proper contrast for visually impaired users.

Usability Metrics

- Reduced number of clicks from 7 to 3.
- Appointment booking time reduced by approximately 50%.
- Improved user satisfaction based on test users.

5. Graphics Issues and Improved Graphics

Graphics Issues in Existing Website

- Outdated design
- Low-quality icons
- Poor color contrast
- Unbalanced layout

Improvements Made

- Modern and clean UI design
- High-quality icons
- Proper spacing and alignment
- Soft and professional healthcare color theme

Screenshot Placement

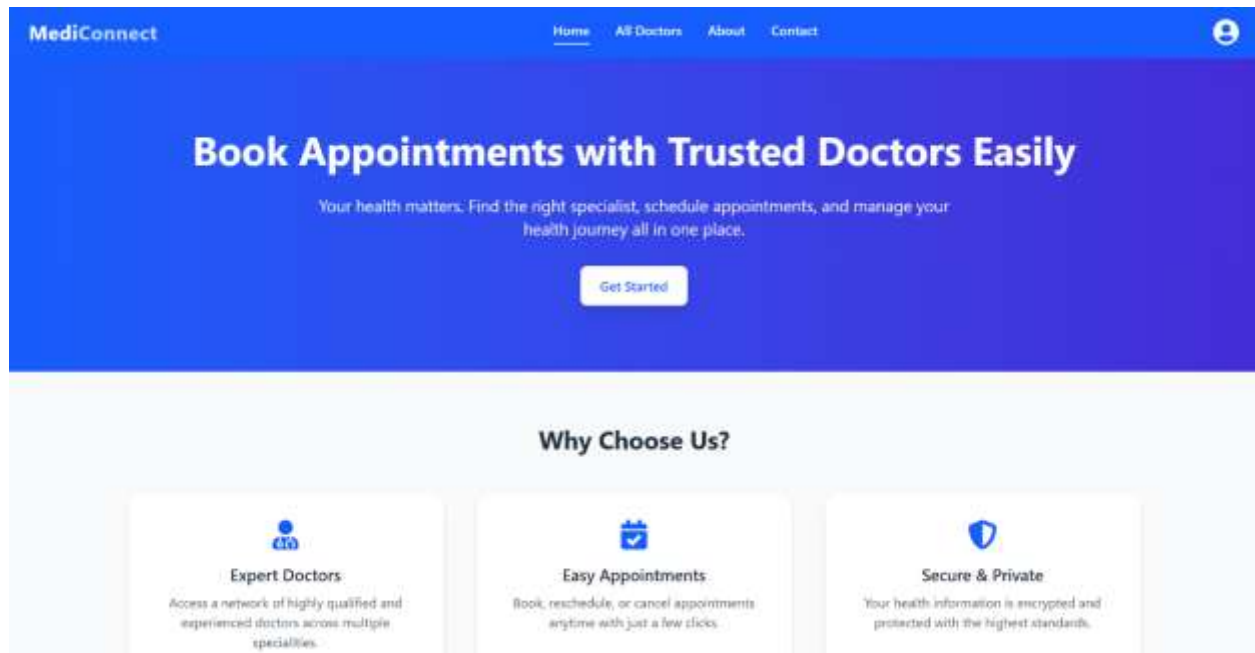


Figure 10 Improved Home page

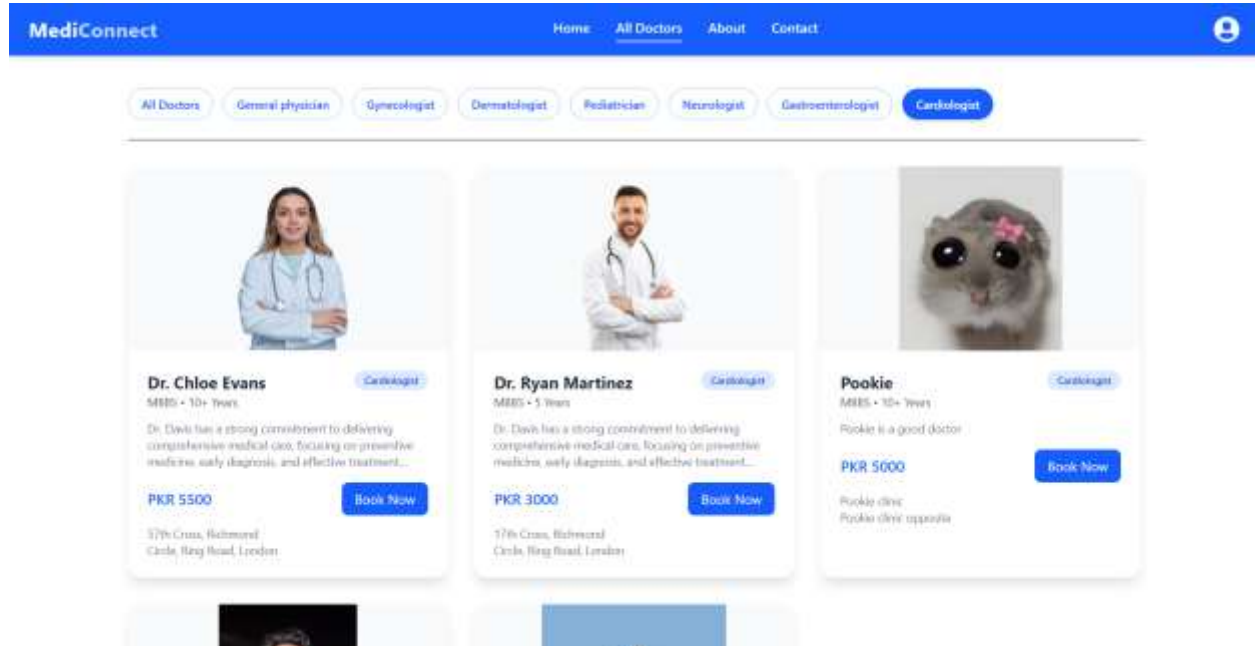


Figure 11 Improved Doctors Page with filtration

6. Web Pages and Backend (Complete Working Prototype)

Frontend Pages

1. Home Page
2. Doctor Listing Page
3. Doctor Profile Page
4. Appointment Booking Page
5. Login / Signup Page
6. Admin Dashboard (Optional)

Backend Features

- User authentication
- Doctor management
- Appointment scheduling
- Database storage
- Confirmation messages

Technologies Used

- Frontend: HTML, CSS, JavaScript / React
- Backend: Node.js / Express
- Database: MongoDB
- Tools: VS Code, Git

The application is fully functional and ready for live demonstration.

7. Work Distribution

Task	Member Name
Usability Analysis on older versiion	Ali Aif
UI/UX Design	Zohaib Ijaz
Frontend Development	Nazish Khan
Backend Development	M Abdul Rafay
Presentation	Nazish Khan
Documentation	M Abdul Rafay

Each member contributed equally and actively to ensure successful project completion.

Conclusion

This project successfully transformed a poorly designed healthcare website into a **user-friendly Doctor Appointment Booking Web Application**. By applying software requirement engineering principles, usability guidelines, and modern design practices, we achieved a functional and efficient system suitable for real-world use.